Exp No: 10	Internetworking with routers in CISCO PACKET TRACER simulator

AIM:

To design and configure internetworks in Cisco Packet Tracer using (a) a router with static IP addresses, (b) a wireless router with a DHCP server and an internet cloud, and (c) a switch, a router and Ethernet cables in the lab.

ALGORITHM / PROCEDURE:

(a) Simple internetwork using a router

1. Design two networks (3 to 4 hosts) and connect them through a router: place one router, two switches and the PCs.
2. Connect each PC to its switch and each switch to a router interface with copper straight-through cables.
3. Assign static IP addresses to the PCs (network 1: 192.168.1.0/24, network 2: 192.168.2.0/24) and set the default gateway of each PC to the router interface address.
4. Open the router CLI and configure both interfaces with `ip address` and `no shutdown`.
5. Ping from a PC of one network to a PC of the other network and trace the routing in simulation mode.

(b) Wireless router, DHCP server and internet cloud

1. Place a wireless router, a cloud, PCs and laptops on the workspace.
2. Connect the wired PC to a LAN port of the wireless router and the router to the cloud; connect the laptops wirelessly.
3. Configure the wireless router (SSID, DHCP enabled) and set the PCs and laptops to obtain the IP address through DHCP.
4. Check the addresses received and test the connectivity.

(c) Inter-network in the lab using switch, router and Ethernet cables

1. Take one router, two switches, the PCs and straight-through UTP cables.
2. Connect the PCs to the switches and the switches to the two Gigabit ports of the router.
3. Assign the IP addresses with the router as the default gateway, configure the router interfaces and ping within the network and between the networks.

Router configuration (Part a and Part c):

Router CLI

```
Router>enable
Router#configure terminal
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
```

DIAGRAM:

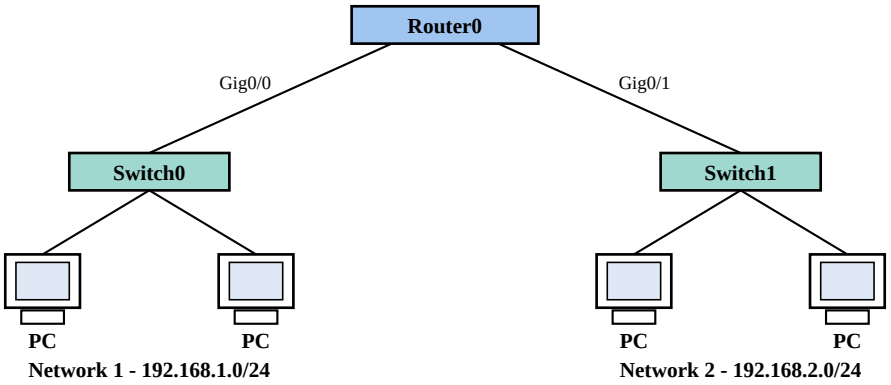
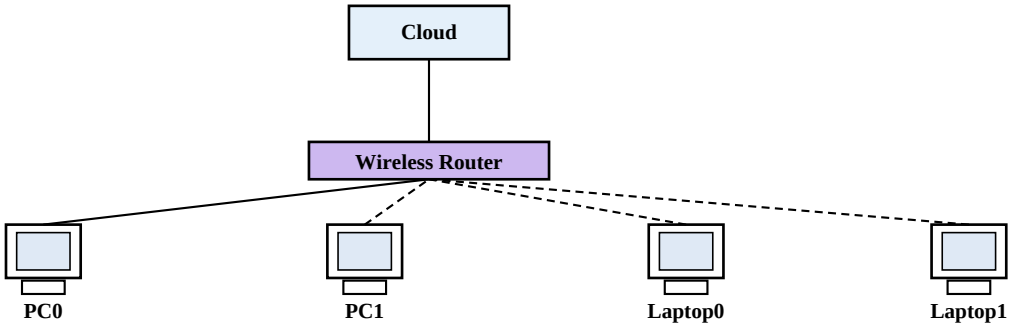


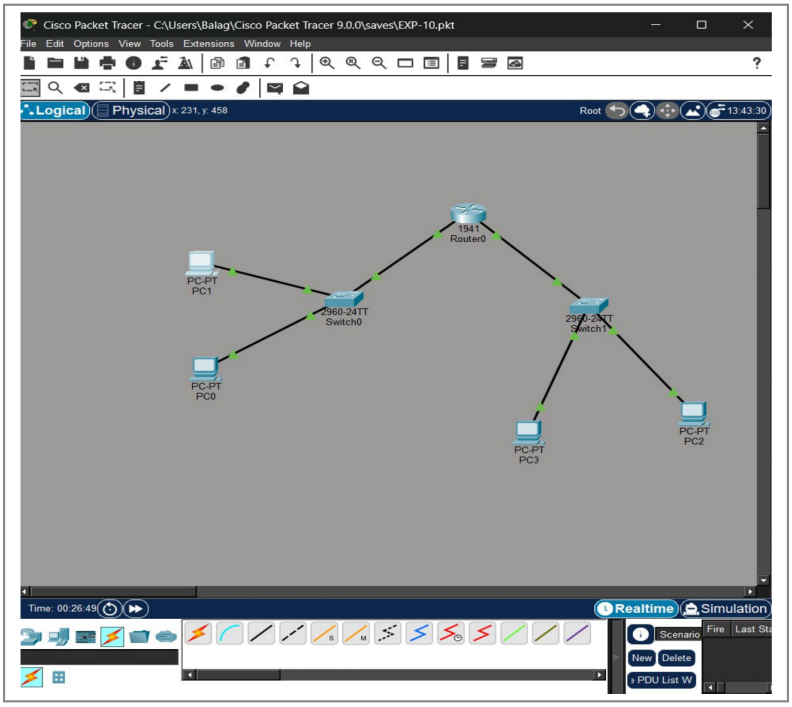
Fig 1: Internetwork of two LANs connected by a router (Part a and Part c)

Fig 2: Wireless router with DHCP and internet cloud (Part b) - dotted lines are wireless links



OUTPUT:

Part (a) - Router with two networks



Packet Tracer topology of Part (a)

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
```

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Router CLI - interfaces configured

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

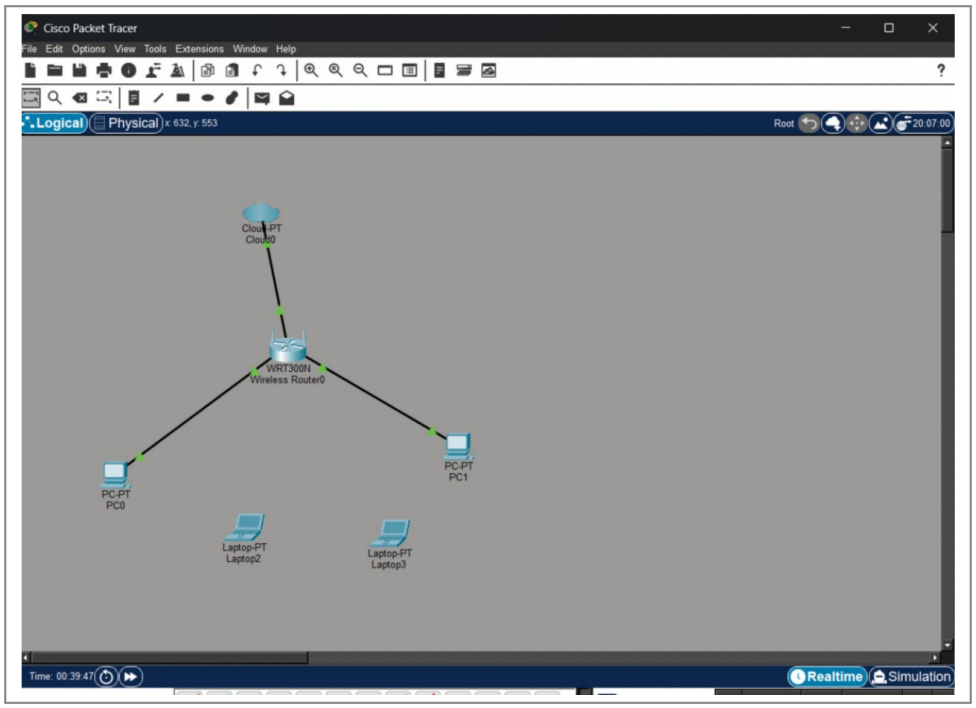
Request timed out.
Reply from 192.168.2.3: bytes=32 time=1ms TTL=127
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

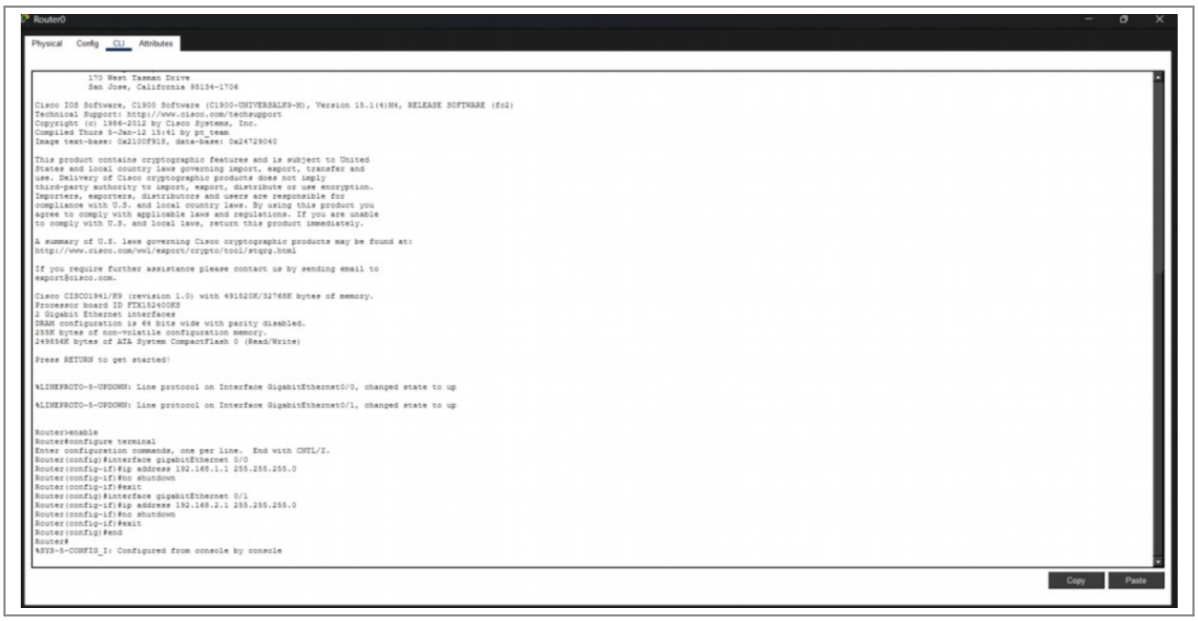
PC1 - ping 192.168.2.3 through the router (3 of 4 replies, TTL = 127)

Part (b) - Wireless router, DHCP and internet cloud

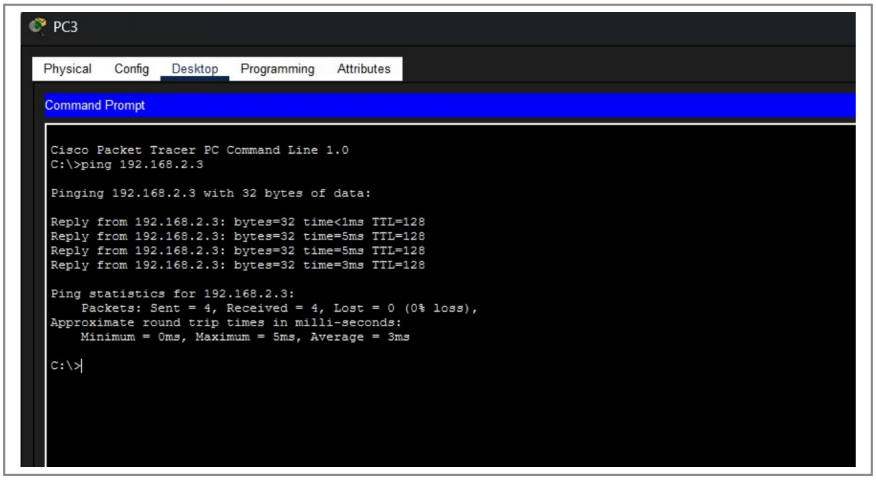


Packet Tracer topology of Part (b)

Part (c) - Lab network with switch, router and Ethernet cables



Router CLI - configuration of the two Gigabit interfaces



PC3 - ping 192.168.2.3 (4 of 4 replies)

In Part (a) the first ping packet is lost because the ARP request has to be resolved. The replies with TTL = 127 show that the packets crossed one router. In Part (c) the ping inside a network gives TTL = 128 because the packets do not pass through the router.

RESULT:

Thus internetworks were designed and configured in Cisco Packet Tracer using a router with static IP addresses, a wireless router with DHCP and a lab network using switches, a router and Ethernet cables. The communication between the networks was verified using the ping command.